

M002 · AND / OR

Topic: AND and OR Conditions · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

New this lesson: **OR** (one is enough). Mix it with **AND**.

- **AND** → runs **only when both** are true.
- **OR** → runs **when at least one** is true.

In Python: **and**, **or**. Detect a wall with `agent.detect(BLOCK, FORWARD)`.

1 · Predict 🧠

```
if wall ahead AND wall on right:
    turn left
```

Wall ahead, but the right is open. Does it turn left? Why?


```
if no wall ahead OR no wall below:
    move forward
```

Ahead is blocked, below is open. Does it move? Why?

2 · Trace It

```
if agent.detect(BLOCK, FORWARD) and agent.detect(BLOCK, RIGHT):  
    agent.turn(LEFT_TURN)  
elif not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

The agent has a wall in front and an open right. What runs?

The agent has open front, wall on right. What runs?

3 · Fill the Blanks

Turn left **only** at a corner (wall front AND wall right).

```
if agent.detect(BLOCK, FORWARD) ____ agent.detect(BLOCK, RIGHT):  
    agent.turn(LEFT_TURN)
```

Move when **either** front OR below is open.

```
if not agent.detect(BLOCK, FORWARD) ____ not agent.detect(BLOCK, DOWN):  
    agent.move(FORWARD, 1)
```

4 · Fix the Bug 🐛

The agent should move when **either** path ahead OR below is clear. This makes it wait for both.

```
if not agent.detect(BLOCK, FORWARD) and not agent.detect(BLOCK, DOWN):  
    agent.move(FORWARD, 1)
```

Write the fixed line:

Why was it wrong?

5 · Show Your Work 📷 🎥

Go to your homework world. Solve the maze using **one AND** and **one OR**.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.